

Scott Grauer-Gray

Email: sgrauer@gmail.com

LinkedIn: <https://www.linkedin.com/in/scott-grauer-gray-30992a1a>

Github: <https://github.com/sgrauer6>

Google Scholar: <https://scholar.google.com/citations?user=y35b43QAAAAJ>

SUMMARY

Masters graduate with research publications, industry and teaching experience, and a record of academic excellence. Interests include program optimization, GPU computing and other hardware accelerators, image processing, and computer vision.

COMPUTER SKILLS

C++, Python, CUDA, Qt, OpenCV, Java, C, OpenGL, OpenCL, Matlab, Visual Studio, SQL, ImageJ, HTML, Git, JIRA

INDUSTRY EXPERIENCE

Software Engineer – 3D Systems *Canfield Scientific*, May 2020-October 2024 (consultant from October 2019-May 2020)

Image processing with focus on 2D and 3D medical images and surfaces using C++. Implemented interface of Psoriasis Assessment Tool for 3D and 2D full body surfaces using Qt. Optimized code to run in parallel on multi-core CPUs.

Senior Software Engineer *Faro Technologies*, December 2015-February 2019

Worked on image processing with Vantage Laser Tracker stereo cameras, including implementation of ActiveSeek features to lock laser on intended target and on navigation to targets that are relatively close together, as well as camera compensation and user interface for camera view. Used C++ in firmware development and Java for user interface. Developed software tool to retrieve exact location of LEDs on a probe using a Mitutoyo microscope in its development environment.

Intern *Intel*, July 2013-July 2014

Worked on augmented reality proof-of-concept application using Intel graphics. Specific work included hole-filling of areas w/ unknown depth from a depth camera and optimization of application across hardware from low-power Intel Atom processor to high-power Intel Core w/ Iris Pro graphics.

Intern *NVIDIA*, May 2011-August 2011

Worked on NPP CUDA library, specifically on optimizing functions used for parallel processing of 1D signals; these functions implemented addition, division, exponent, square root, and arctan operations on each signal element.

EDUCATION

Masters in Computer and Information Sciences (CISC), University of Delaware, Newark, DE

Bachelor of Science in Computer Science, minor in math, University of Massachusetts, Amherst, MA

SELECTED PUBLICATIONS

- S. Grauer-Gray, W. Killian, R. Searles, J. Cavazos. Accelerating Financial Applications on the GPU. Sixth Workshop on General Purpose Processing using GPUs, 2013.
- S. Grauer-Gray, L. Xu, R. Searles, S. Ayalasomayajula, J. Cavazos. Auto-Tuning a High-Level Language Targeted to GPU Codes. Proceedings of Innovative Parallel Computing (InPar), 2012.
- S. Grauer-Gray, J. Cavazos. Optimizing and Auto-Tuning Belief Propagation on the GPU . In 23rd International Workshop in Languages and Compilers for Parallel Computing (LCPC), 2010.
- S. Grauer-Gray, C. Kambhamettu, K. Palaniappan. GPU Implementation of Belief Propagation Using CUDA for Cloud Tracking and Reconstruction. In 5th IAPR Workshop on Pattern Recognition in Remote Sensing (PRRS), 2008.